

Machine Learning Scenario Based Answer

1.A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.

Q: Identify the problem type and outline the step-by-step logic to solve it.

House Price Prediction

Problem Type: Supervised Learning – Regression

Logic:

1. Collect house data.
2. Clean and preprocess data.
3. Select features (square footage, bedrooms, location).
4. Split into train and test data.
5. Train a regression model.
6. Predict house prices.
7. Evaluate using R^2 Score.

2.A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Fraud Transaction Detection

Problem Type: Supervised Learning – Classification

Logic:

1. Collect transaction history.
2. Label transactions as fraud/not fraud.
3. Preprocess data.
4. Train classification model.
5. Predict fraudulent transactions.
6. Evaluate accuracy.

3.A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Customer Segmentation

Problem Type: Unsupervised Learning – Clustering

Logic:

1. Collect shopping data.
2. Clean data.
3. Select customer behavior features.
4. Apply clustering (K-Means, DBSCAN, Algomerative).
5. Group similar customers.
6. Use groups for promotions.

4.A company wants to estimate an employee's salary based on their years of experience, job title, and education level.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Employee Salary Prediction

Problem Type: Supervised Learning – Regression

Logic:

1. Collect employee details.
2. Preprocess data.
3. Select features (experience, job title, education).
4. Train regression model.
5. Predict salary.
6. Evaluate model.

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Spam Email Detection

Problem Type: Supervised Learning – Classification

Logic:

1. Collect emails.
2. Label spam and non-spam.
3. Convert text into numerical features.
4. Train classification model.
5. Predict spam or not spam.
6. Measure accuracy.

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Sentiment Analysis

Problem Type: Supervised Learning – Classification

Logic:

1. Collect customer reviews.
2. Label positive and negative reviews.
3. Process text data.
4. Train classification model.
5. Predict sentiment.
6. Evaluate results.

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Insurance Claim Prediction

Problem Type: Supervised Learning – Classification

Logic:

1. Collect customer details.
2. Preprocess data.
3. Train classification model.
4. Predict claim likelihood.
5. Evaluate accuracy.

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Movie Recommendation by Grouping Users

Problem Type: Unsupervised Learning – Clustering

Logic:

1. Collect viewing history.
2. Analyze user preferences.
3. Cluster similar users.
4. Recommend movies preferred by the same cluster.
5. Improve recommendations.

9.A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Patient Recovery Time Prediction

Problem Type: Supervised Learning – Regression

Logic:

1. Collect patient information.
2. Preprocess data.
3. Train regression model.
4. Predict recovery time.
5. Evaluate using R^2 Score.

10.A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Student Final Exam Score Prediction

Problem Type: Supervised Learning – Regression

Logic:

1. Collect student data.
2. Select features (study hours, attendance, past marks).
3. Train regression model.
4. Predict final score.
5. Evaluate performance.

